

Professional Experience

Artificial Intelligence Engineer, Wakeb_data, Cairo, Egypt

Jan 2025 - Present

- Built a self-hosted, end-to-end, prompt-based AI presentation generation tool that produces full decks from a single prompt; integrated PDF parsing, web search, and RAG for grounded content, LLM structured output for slide mapping, and a routing agent that decides per slide whether to reuse document images, search online, or generate new ones via VLMs.
- Built and launched a full Agentic Process Automation (APA) system enabling users to build complete executable workflows through chat; orchestrated LLM agents with LangGraph and stateful checkpointing, integrating external services (Salla, WhatsApp, Slack), significantly advancing beyond traditional rule-based RPA.
- Built a production RAG pipeline with dense vector retrieval over pgvector and Qdrant, enabling semantic search and factual grounding for LLM-generated outputs.
- Delivered end-to-end computer vision and geospatial analytics systems, including satellite image segmentation, object detection, and change detection for land cover mapping and environmental monitoring.
- Fine-tuned open-source Vision-Language Models and foundational models for domain-specific remote-sensing tasks, and optimized them for production deployment on edge devices (NVIDIA Jetson) with Docker.
- Led the annotation team: gathered requirements, delegated and reviewed tasks for accuracy, and automated data collection, preprocessing, and validation to deliver model-ready datasets.

Keywords: LangGraph, LangChain, LLMs, RAG, Agentic AI, Prompt Engineering, Structured Output, VLMs, pgvector, Qdrant, TensorFlow, PyTorch, Docker, NVIDIA Jetson, Open Vocabulary, Domain Adaptation, MLflow, Alembic, GitHub Actions, Nginx, Ollama

Artificial Intelligence Engineer, Medsoft, Remote

Jan 2023 - Jan 2025

- Built advanced 3D segmentation models in TensorFlow to enhance accuracy in medical imaging; trained and evaluated large-scale models on **AWS SageMaker** with S3 storage.
- Applied **Statistical Shape Modeling** combined with deep autoencoders (VAEs, GANs) for generative reconstruction and synthetic data creation.
- Implemented an innovative 3D landmarking approach using a 2D U-Net model, integrating computer graphics with deep learning to reduce training overhead.
- Collaborated closely with international teams, including the **DePuy Synthes R&D team at Johnson & Johnson**, to deliver innovative solutions in medical imaging and reconstruction.

Keywords: TensorFlow, PyTorch, AWS SageMaker, S3, UNet, GANs, VAEs, Autoencoder, Statistical Shape Modeling

Computer Vision Engineer, DevisionX, Remote

Nov 2022 - Jan 2023

- Developed an automated machine learning pipeline for image classification, object detection, and segmentation using TensorFlow AutoML, streamlining model training and benchmarking.

Keywords: Auto ML, Keras-tuner, YOLO, Benchmarking, TensorFlow, PyTorch.

Projects

mini-RAG, *project page*

Developed a production-ready **Retrieval-Augmented Generation (RAG)** application implementing dense vector retrieval via pgvector and hybrid storage with MongoDB. Focused on chunking strategy, retrieval accuracy, and API scalability. Deployed with Docker Compose, monitored via Prometheus and Grafana, with Nginx as reverse proxy.

Keywords: RAG, Semantic Search, pgvector, MongoDB, FastAPI, Docker, Prometheus, Grafana, Nginx, RESTful API, PostgreSQL

Needle - Necessary Elements for Deep Learning, *project page*

A deep learning framework built from scratch in Python and C++, based on automatic differentiation (PyTorch-style). Covers tensor ops, backprop, and neural network primitives — demonstrating systems-level understanding of ML internals.

Keywords: Automatic Differentiation, Tensor, Python, C++, OOP

Kinematics Identification Using Deep Learning, *project page*

A deep learning toolkit for automated swallowing kinematic analysis in X-Ray videofluoroscopy. Combined action recognition, object tracking, and image segmentation (U-Net, SSD, VGG16) into a computer-aided clinical system. Published work.

Keywords: Action Recognition, Object Tracking, Image Segmentation, U-Net, SSD, VGG16

Skills & Technologies

- **LLM & Agentic AI:** LangChain, LLMs, RAG Architectures, Prompt Engineering, Agentic Pipelines, Vector Search, Semantic Search, Ollama.
- **ML/DL & MLOps:** TensorFlow, PyTorch, Hugging Face, Transformers, GANs, VAEs, NLTK, MLOps, Vision-Language Models, Differential Geometry, Statistical Shape Modeling.
- **Computer Vision:** Satellite Image Segmentation, Object Detection, Change Detection, Action Recognition, Object Tracking, Image Segmentation.
- **DevOps & Build Systems:** Docker, Docker Compose, Prometheus, Grafana, Nginx, Git/GitHub, GitHub Actions, Alembic, Make.
- **Cloud Computing:** AWS S3, AWS SageMaker.
- **Databases & Web Dev:** MongoDB, PostgreSQL with pgvector, Qdrant, Flask, HTML, CSS, MySQL, FastAPI, RESTful API.
- **Programming:** Python, C++, Scala.
- **Computer Graphics:** Libigl, Scalismo, VTK, CGAL.
- **Other Tools:** NVIDIA Jetson, Ollama, $\text{\LaTeX}$, Markdown, Jupyter, VSCode.

Publications

[P1] AI-Powered Toolkit for Automated Swallowing Kinematic Analysis in X-Ray Videofluoroscopy

Education

Cairo University Faculty of Engineering | Bachelors in **Biomedical Engineering**

2017-2022